Southern Cross University

The Endocrine System

Medicines for bone disorders

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The endocrine system: Medicines for bone disorders

COMMONWEALTH OF AUSTRALIA

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Objectives for Endocrine System: Bone Disorders

- Explain the effects of parathyroid hormone and calcitonin and their role in bone health.
- Explain the process of bone formation and reformation
- Using two specific examples (bisphosphonates and teriparatide), explain how pharmaceutical agents can be beneficial for people with bone loss. Explain the main adverse drug reactions of these medications.

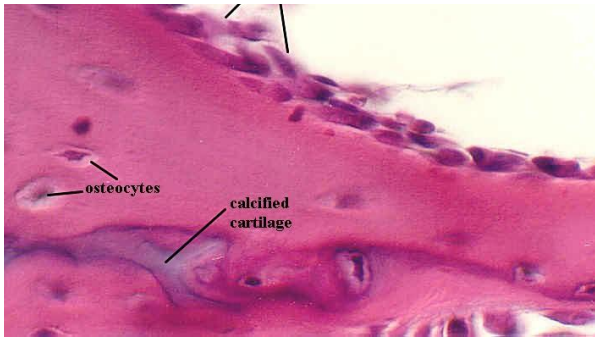
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Medications of the Endocrine System: Medications for bone disorders

Lesson 1:

Understanding Bone Health

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Bone health: Key cells and hormones

The balance between osteoblast and osteoclast activity is important

- Osteoclasts
  - Digests the bone
  - Secrete cytokines that stimulate resorption
  - One small section of bone at a time
- Osteoblasts rebuild the bone
- Bone formation and breakdown involves participation of the hormones from the parathyroid gland and the hormone commonly referred to as vitamin D

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Factors that contribute to osteoporosis

Constitutional	Disease & drugs	Lifestyle
Female	hyperthyroidism	Smoking
Age	hyperparathyroidism	Physical inactivity
Post menopause	Cushing's	Low calcium and vitamin D intake
Slight body build	Malabsorption syndromes.	
Late menarche	Corticosteroids	
	Androgen deprivation therapy	
	Glitazones	
	(Drugs that increase the falls risk)	

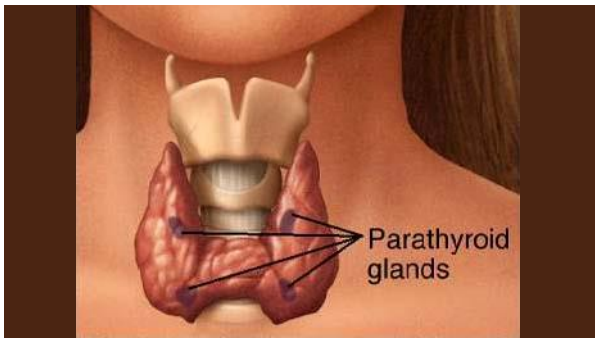
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Lesson 2:

Understanding Bone Health

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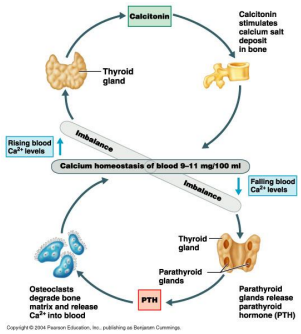
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PTH: The hormone of the parathyroid gland

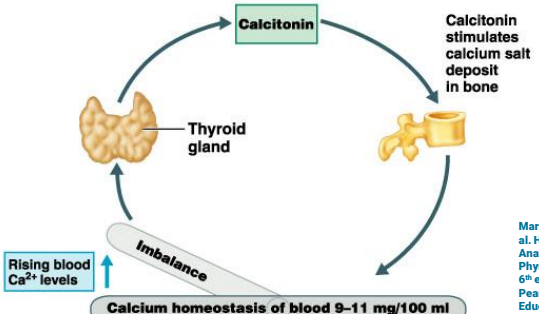
Parathyroid Hormone (PTH)
- Increased release in response to decreased serum calcium levels
- Increases osteoclastic activity
- Liberates free calcium
- Decreases excretion of calcium by the kidneys

Calcitonin is released from the thyroid gland
- Responds to high serum calcium
- Inhibits osteoclastic activity and formation

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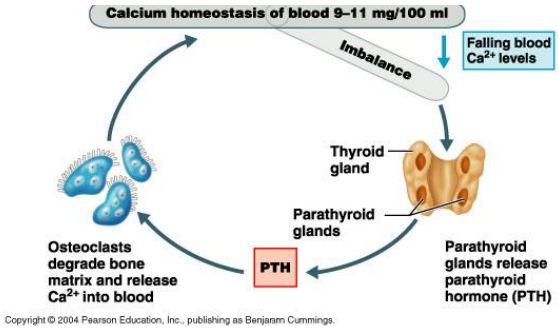


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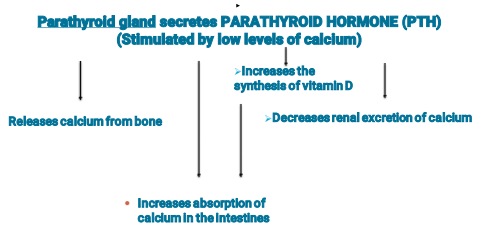


Marieb EN et al. Human Anatomy and Physiology 6th ed. Pearson Education

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Important bone disorders

Osteoporosis
- Decreased bone mass and density

Osteomalacia and Rickets
- Softening of the bones due to decreased mineralisation
- Most commonly caused by inadequate Vitamin D

Osteonecrosis
- Cellular death of bone cells due to inadequate blood supply

Paget's disease
- Abnormal enlargement and weakening of the bone

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Glucocorticoid induced bone disease

- Leading cause of secondary & iatrogenic osteoporosis.
- Lumbar spine & proximal femur.
- 2 phases:
  - 6-12% loss in first year.
  - 3% loss pa thereafter.
  - Rapid increase in risk of hip and vertebral fracture.
  - Fracture risk increases by 75% in first 3 months of treatment before a decline in BMD.
- *Osteonecrosis* may be as high as 40% of patients
  - Osteonecrosis is death of bone due to decreased blood supply

NEJM 2011; 365:62-70

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Prevention of osteoporosis

Calcium and vitamin D supplements.
- *Reduce rate of bone loss.*
- *Vitamin D 1000iu/day.*

No calcium supplements after about 50 y/o due to increased risk of MI

Exercise, especially weight bearing exercise before the age of 30

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Lesson 3:

Bisphosphonates, Teriparatide and SERMs

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Treatment of osteoporosis

First Line:

- Oral Bisphosphonates
  - Inhibit bone breakdown and resorption
  - Selective localisation and prolonged storage in bone
  - Increase BMI in the *femoral neck* and *lumbar spine*.
- Osteonecrosis of the jaw.

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Oral bisphosphonates

(The first 's' is normally silent)

Oral bisphosphonates end in …dronate:

- Alendronate *(Fosamax)*
- Risedronate *(Actonel)*
- Pamidronate *(Aredia)*
- Zoledronate *(Zometa, Aclasta)*

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Other treatments for osteoporosis

- Intravenous bisphosphonates
  - IV Zoledronic acid
    - Once yearly infusion
    - Rapid skeletal protection



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Other treatments for osteoporosis

- Selective oestrogen receptor modulators (SERMs)
  - …oxifene medications like raloxifene, tamoxifen, and lasofoxifene
    - Agonist at the bone (acts like oestrogen at the bone
    - Antagonist at the breast, ovary (blocks the oestrogen in the areas associated with oestrogen related cancers)

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Other treatments for osteoporosis

- Teriparatide, the only anabolic treatment
  - Recombinant human parathyroid hormone; identical to a portion of human PTH
  - Intermittent use activates osteoblasts more than osteoclasts
  - It is effective in growing bone (8% increase in bone density in the spine after one year)
  - Daily s/c injection
  - Osteosarcoma??
  - Hypercalcaemia.



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References

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